higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T1090**(E)**(A10)T

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N6

(8090086)

10 April 2019 (X-Paper)
09:00–13:00

OPEN-BOOK EXAMINATION

This question paper consists of 4 pages and 1 diagram sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MECHANICAL DRAWING AND DESIGN N6
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL calculations must have at least THREE steps, namely the formula, substitution and answer.
5. Use g = 9,81 m/s^2.
6. Write neatly and legibly.

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QUESTION 1

A cam with a radial flat-face follower is fitted with a spring as shown in FIGURE 1 on the DIAGRAM SHEET (attached). The cam is mounted on a solid shaft and overhang the journal bearing by 200 mm.

CAM INFORMATION

The spring force is 1 200 N.
The frictional force between the guide and the follower is 180 N.
The coefficient of friction between the cam and the follower is 0,12.
The rise is uniform at a rotation of 60° and the follower moves 10 mm.

BELT INFORMATION

The coefficient of friction between the belts and the pulley is 0,45.
The pulley effective diameter is 400 mm.
The angle of contact between the belts and the pulley is 160°.
The mass of the pulley is 40 kg.

JOURNAL BEARING INFORMATION

The coefficient of friction between the bearing and the shaft is 0,08.
The rotational frequency of the shaft is 1 000 r/min.
The bearing pressure is 39,166 kPa.
The specific heat capacity of the cooling oil is 3,187 kJ/kg.°C.
The rise in temperature of the cooling water is 8 °C.
The rate of flow of the cooling oil is 0,064 ℓ.

BEARING INFORMATION

Both bearings are journal bearings.
The coefficient of friction between the bearing and the shaft is 0,08.

CLUTCH INFORMATION

The centrifugal clutch has FOUR shoes which slide radially on a spider key to the driving shaft.
The coefficient of friction between the shoes and the clutch drum is 0,25.
The clutch engages at 400 r/min.
The mass of each shoe is 3,2 kg.

Calculate from the given information the:

1.1 Frictional force between the follower and the cam (2)

1.2 Torque provided by the cam shaft (11)

1.3 Maximum bending on the shaft induced by the cam (4)

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1.4 Safe diameter of the cam shaft if the maximum bending and shear stresses are 50 MPa and 45 MPa respectively (9)

1.5 Tension in the slack and tight side (5)

1.6 Length of the shaft supporting the pulley from the centre bearing (assume that the shaft has a maximum bending moment at the centre of the bearing and is also in a state of equilibrium) (5)

1.7 Resultant reaction on the bearing (4)

1.8 Journal bearing length (6)

1.9 Specific gravity of cooling liquid on the journal bearing (8)
[54]

QUESTION 2

FIGURE 2 on the DIAGRAM SHEET (attached) shows a four-speed gearbox for a motor vehicle. The pinion A transmits 55 kW at 1 500 r/min. The number of teeth for gears A, B, D, E and H is 31, 64, 53, 37 and 67 respectively. ALL gears have the same module and are made of cast steel with a basic stress of 140 MPa. The pressure angle for ALL gears is 20°.

2.1 Calculate:

2.1.1 The FOUR different output speeds of the gearbox (15)

2.1.2 If there is no interference between gears G and H (7)

2.1.3 The face width for gears E and F (15)

2.2 If a gearbox is coupled to a single-plate clutch to give the gearbox a smooth gear change, the clutch is fitted between the engine and the gearbox. The inside diameter of the clutch plate is two-thirds of the outside diameter. The uniform pressure on the contact surface is 200 kPa. The coefficient of friction 0,45.

Assume uniform pressure and calculate the inside and outside diameters of the clutch plate. (9)
[46]

TOTAL: 100

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DIAGRAM SHEET

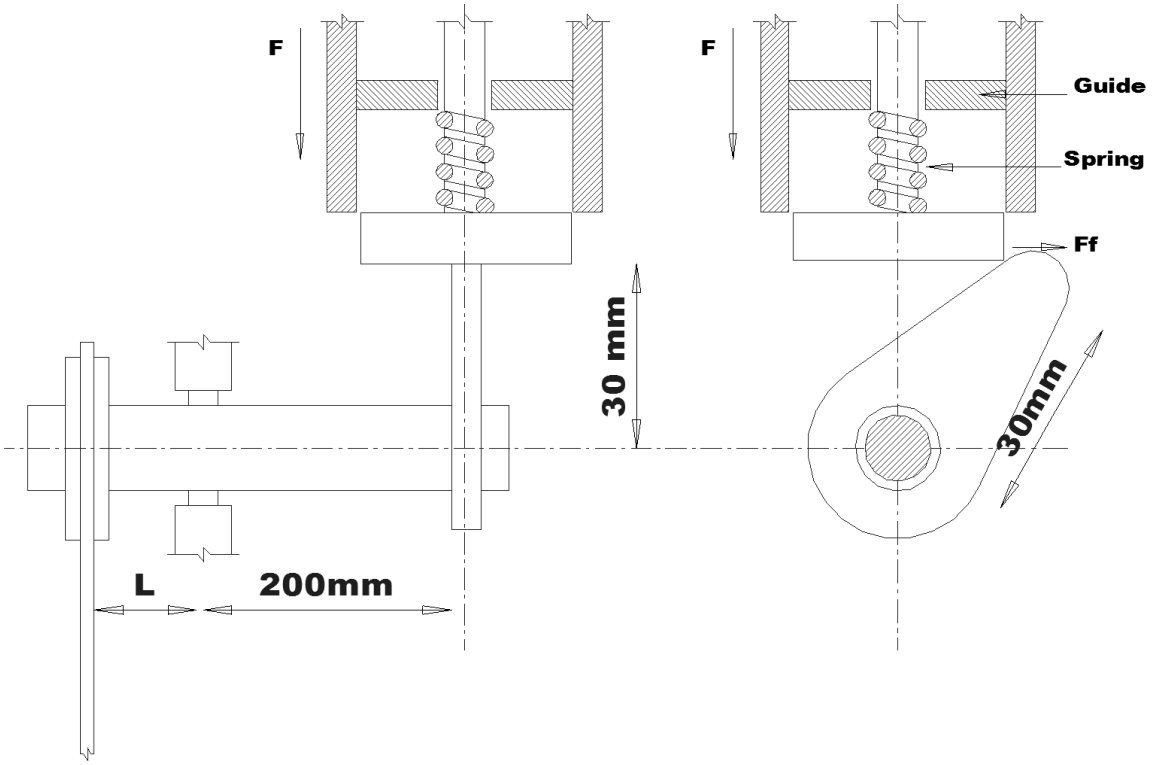


FIGURE 1

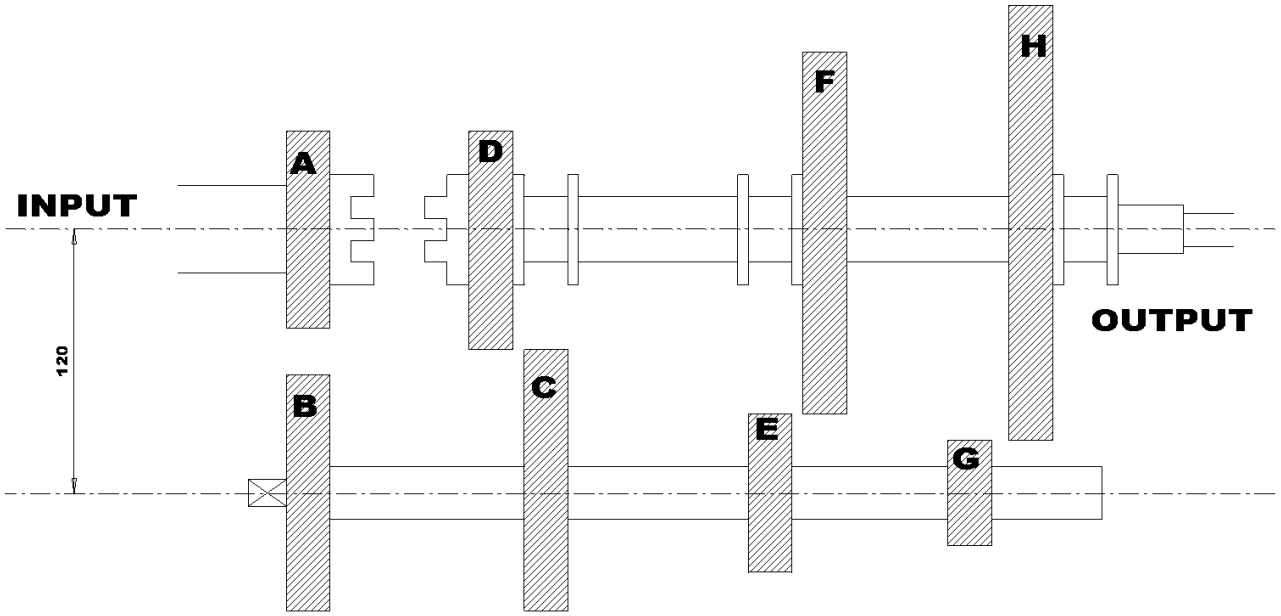


FIGURE 2

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